

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

NOV 2013

CL 208 Advance Surveying

Date: 12.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define following

1) Mistakes 2) Indirect Observation 3) True Value 4) Most probable error 5) Residual error 6) Observed value 7) Triangulation

[07]

Q - 2 (a) Find most probable value for following angle.

[06]

AOB $85^{\circ} 42' 28''.75$ wt. 3

BOC $102^{\circ} 15' 43''.26$ wt. 2

COD $94^{\circ} 38' 27''.22$ wt. 4

DOA $79^{\circ} 23' 24''.77$ wt. 2

- (b) To find elevation of the top(Q) of a hill, a flag staff of 2m height was erected and observations were made from two stations P and R, 60 m apart. The horizontal angle measured at P between R and the top of the flag staff was $60^{\circ} 30'$ and that measured at R between the top of the flag staff and P was $68^{\circ} 18'$. The angle of elevation to the top of the flag staff was measured to be $10^{\circ} 12'$ at P. The angle of elevation to the top of the flag staff was measured to be $10^{\circ} 48'$ at R. Staff reading on B.M. When the instrument at P=1.965m and that with the instrument at R=2.055m. Calculate the elevation the top of the hill if that of B.M was 435.065m

[08]

OR

Q - 2 (a) Explain method of correlates for finding most probable value.

[06]

- (b) Calculate distance and height formula for trigonometric leveling when Base of the object inaccessible: Instrument stations in the same vertical plane with elevated object, Instrument axis at same level.

[08]

Q - 3 (a) To determine the elevation of station P in a tacheometer survey, the following observations were made with staff held vertical. The instrument was fitted with an anallactic lens and its multiplying constant was 100. If R.L of BM is 250m calculate R.L. of P

[06]

Instrument st.	H.I	Staff st.	Vertical angle	Staff readings
O	1.45	B.M.	-6°	1.335, 1.895, 2.460
O	1.45	C.P.	$+8^{\circ} 30'$	0.780, 1.265, 1.745
P	1.40	C.P.	$-6^{\circ} 30'$	1.155, 1.615, 2.075

- (b) Give difference between fixed hair method and movable hair method. [06]
 (c) Define : 1) Signals 2) Phase of signal [02]

OR

- Q - 3 (a) Explain field procedure to find tachometric constant. [06]
 (b) Enlist different cases for tangential tacheometry with neat sketch. Give distance and elevation formula for any one case. [06]
 (c) Define : 1) Well conditioned Triangle 2) Base line [02]

SECTION - II

- Q - 4 (a) Write important points for selection of triangulation station. [05]
 (b) Define 1) Remote sensing 2) GPS [02]
 Q - 5 (a) Enlist different applications of Remote sensing. Explain any two in details. [07]
 (b) Define [07]
 1) Celestial sphere 2) Azimuth 3) Vertical circle 4) Hour circle 5) Zenith 6) Nadir
 7) Altitude

Or

- Q - 5 (a) Explain effect of EM radiation on Atmosphere. [07]
 (b) Determine the azimuth and altitude of a star from the following data [07]
 i) Declination of star = $20^{\circ} 30' N$ ii) Hour angle of star = $42^{\circ} 6'$
 iii) Latitude of observer = $50^{\circ} N$

- Q - 6 (a) Define following [06]
 1) Isocentre 2) Relief displacement 3) Principal Point
 (b) Two points A and B having elevations of 500 m and 300 m respectively above datum appear on the vertical photographs having focal length of 20 cm and flying altitude of 2500 m above datum. Their corrected photographic co-ordinates are as follows.

Point	Photographic Co-ordinates	
	x (cm)	y (cm)
A	+2.65	+1.36
B	-1.92	+3.65

Or

- Q - 6 (a) Explain scale of vertical photographs with neat sketch. [06]
 (b) The scale of an aerial photograph is 1 cm = 100 m. The photograph size is 20 cm X 20 cm. Determine the number of photographs required to cover an area of 100 sq km if the longitudinal lap is 60% and the side lap is 30%. [08]
